SUMMARY

- A ray of light gives the direction of light.
- Law of reflection is, $i = r$
- Paraxial rays are the rays travelling close to the principal axis of the mirror and make
- small angles with it.
- The relation between focal length and radius of curvature in spherical mirror is, $2f = R$ (or) $f = \frac{R}{2}$
- Cartesian sign conventions are to be followed to trace image formed by spherical mirrors.
- The mirror equation is, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$
- The magnification in spherical mirror is, $m = \frac{h'}{h} = -\frac{v}{u}$, $m = \frac{h'}{h} = \frac{f - v}{f} = \frac{f}{f - u}$
- Light travels with lesser velocity in optically denser medium.
- Refractive index is the ratio of speed of light in vacuum to speed of light in medium, $n = \frac{c}{v}$
- Optical path is the equivalent path travelled in vacuum in the same time light travels through a optically denser medium. $d' = nd$
- Law of refraction also called as Snell's law in ratio form is, $\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$.
- In product form is, $n_1 \sin i = n_2 \sin r$
- The relative refractive index of second medium with respect to first medium is, $n_{21} = \frac{n_2}{n_1}$
- The apparent depth is always lesser than actual depth. The equation for apparent depth is, $d' = \frac{d}{n}$
- The critical angle of incidence i_c for a ray incident from a denser to rarer medium, is that angle for which the angle of refraction is 90°. For $i > i_c$, total internal reflection occurs.
- Equations for critical angle incidence is, $\sin i_c = \frac{1}{n}$ (or) $i_c = \sin^{-1}\left(\frac{1}{n}\right)$
- Snell's window is the restricted area of circular illumination which appears when seen from water due to critical angle incidence. The radius of the circular illumination is, $R = d\left(\frac{1}{\sqrt{n^2 - 1}}\right)$ (or) $R = \frac{d}{\sqrt{n^2 - 1}}$
- Optical fibre makes use of critical angle incidence. The acceptance angle in optical fibre is, $i_a = \sin^{-1}\left(\sqrt{\frac{n_1^2 - n_2^2}{n_3^2}}\right)$. Here, n_1, n_2, n_3 are the refractive indices of core, cladding and surrounding medium respectively.
- Glass slabs produce a lateral displacement on the light falling on it. The equation for lateral shift is, $L = t\left(\frac{\sin(i - r)}{\cos(r)}\right)$

XII_Physics_Vol-2_UNIT-6_EM.indd 45 26-12-2022 11:38:43